THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS CHAPTER TEST SET B

Chapter 4: **Exploring Algebraic Identities** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Expand $(2x + 5)^2$: 1

(a) $4x^2 + 20x + 25$ **(b)** $4x^2 + 25$ **(c)** $4x^2 + 20x - 25$ **(d)** $4x^2 + 10x + 25$

2. Expand $(2x - 3)^2$: 1

(a) $4x^2 + 12x + 9$ **(b)** $4x^2 - 6x + 9$ **(c)** $4x^2 - 12x + 9$ **(d)** $4x^2 - 9$

3. The value of 29^2 using a suitable identity is: 1

(a) 831 **(b)** 941 **(c)** 841 **(d)** 861

4. The expression $4s^2 + 20st + 25t^2$ can be factorised as: 1

(a) $(4s + 5t)^2$ **(b)** $(2s - 5t)^2$ **(c)** $(2s + 25t)^2$ **(d)** $(2s + 5t)^2$

5. $(x + 5)(x - 2)$ equals: 1

(a) $x^2 + 3x - 10$ **(b)** $x^2 + 3x + 10$ **(c)** $x^2 - 10x + 3$ **(d)** $x^2 + 7x - 10$

6. Factorise $x^2 - 5x + 6$: 1

(a) $(x - 2)(x + 3)$ **(b)** $(x - 3)(x + 2)$ **(c)** $(x + 2)(x + 3)$ **(d)** $(x - 3)(x - 2)$

7. $(7x + 2)(7x - 2)$ equals: 1

(a) $49x^2 - 4$ **(b)** $49x^2 - 28x + 4$ **(c)** $49x^2 + 28x + 4$ **(d)** $49x^2$

8. $(2x - 1)^3$ equals: 1

(a) $8x^3 - 6x^2 + 3x - 1$ **(b)** $8x^3 - 1$ **(c)** $8x^3 + 12x^2 + 6x + 1$ **(d)** $8x^3 - 12x^2 + 6x - 1$

9. $(x - y)(x^2 + xy + y^2)$ equals: 1

(a) $x^3 + y^3$ **(b)** $x^2 - y^2$ **(c)** $(x - y)^3$ **(d)** $x^3 - y^3$

10. The three consecutive squares 25, 36, 49: (smallest + largest) − 2 × middle equals: 1

(a) 0 **(b)** 1 **(c)** 4 **(d)** 2

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Factorise $9x^2 + 24xy + 16y^2$. 2

12. Factorise $4s^2 + 20st + 25t^2$. 2

13. Factorise $16y^2 - 24y + 9$. 2

14. Factorise $49x^2 + 14xy + y^2$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. What is the side of the cube whose volume is $p^3 + 6p^2q + 12pq^2 + 8q^3$? 3

16. Simplify $\dfrac{x^2 - 7x + 12}{5x^2 + 5x - 100}$, assuming the denominator is non-zero. 3

17. Simplify $\dfrac{36s^2 - 12st + t^2}{t^2 + 2ts - 48s^2}$ (denominator $\neq 0$). 3

OR

Simplify $\dfrac{p^4 - 16}{p^2 - 4p + 4}$ ($p \neq 2$). 3

18. Using $x^3 + y^3 + z^3 - 3xyz = (x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx)$ show that if $x + y + z = 0$ then $x^3 + y^3 + z^3 = 3xyz$. Evaluate $(-5)^3 + 2^3 + 3^3$. 3

OR

The sum of three numbers is 10 and the sum of their squares is 38. Find $xy + yz + zx$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Expand $(x + 2)^3$. 5

(ii) Factorise $x^3 - 8$.

(iii) Evaluate 103^3 using $(100 + 3)^3$.

OR

The sum of three numbers is 10, the sum of their squares is 38 and their product is 25. 5

(i) Find $xy + yz + zx$.

(ii) Hence find the sum of their cubes.

20. Simplify 5

(i) $\dfrac{3p^2 - 3pq - 18q^2}{p^2 + 3pq - 10q^2}$ and

(ii) $\dfrac{4y^2 - 20yz + 25z^2}{25z^2 - 4y^2}$ (denominators non-zero).

OR

Case study: Śhrīdharāchārya (750 CE) gave $a^2 = (a + b)(a - b) + b^2$ for quick squaring. 5

(i) Find 35^2, 65^2 and 85^2 with $b = 5$.

(ii) Observe the pattern in the units and tens of these squares.

(iii) Find 105^2 using the same method.

END OF QUESTION PAPER — Total: 40 Marks

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